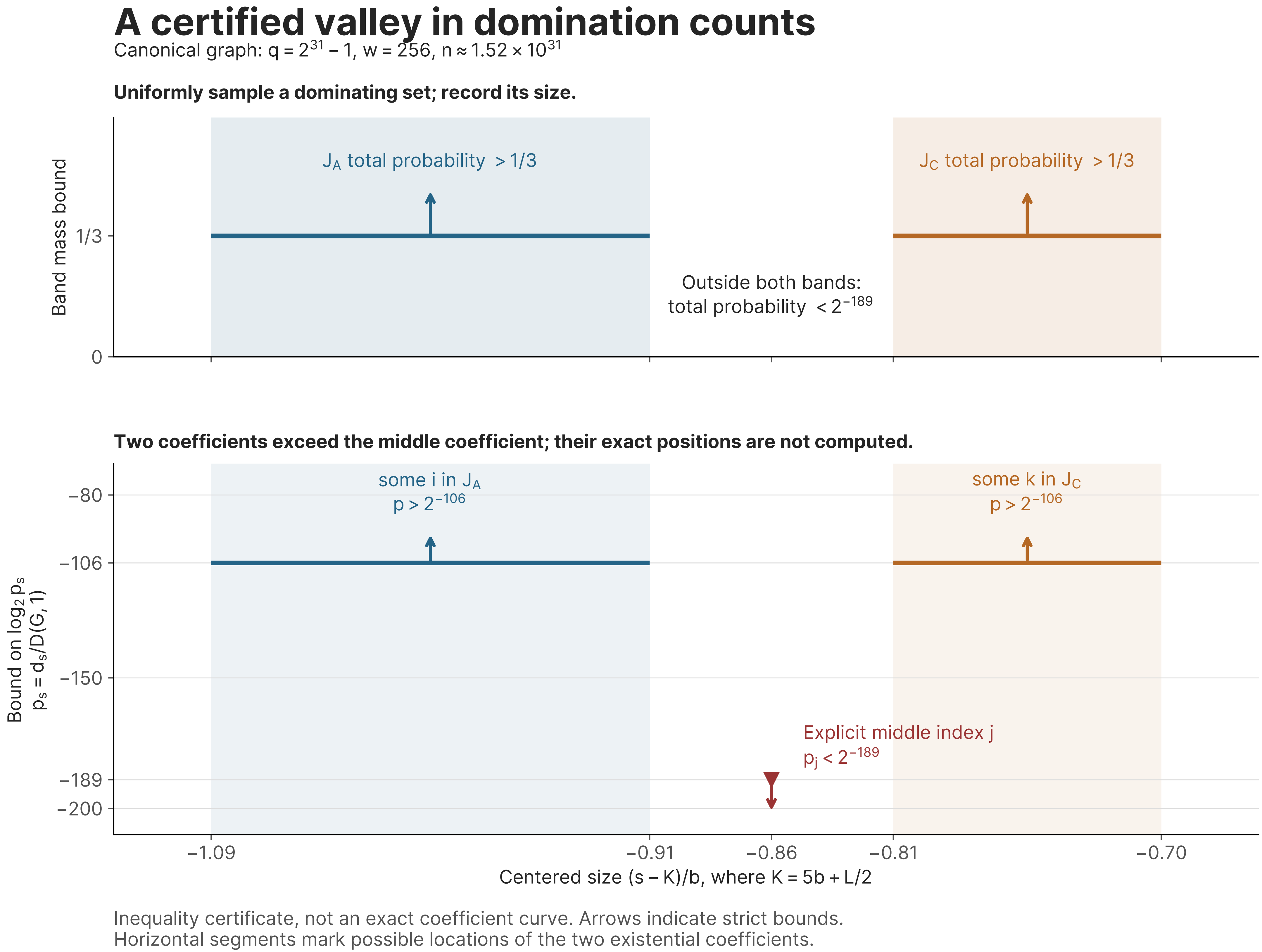

A certified valley in domination counts
Canonical graph: $q = 2^{31} - 1$, $w = 256$, $n \approx 1.52 \times 10^{31}$
Uniformly sample a dominating set; record its size.
Band mass bound
1/3
0
J_A total probability $> 1/3$
J_C total probability $> 1/3$
Outside both bands:
total probability $< 2^{-189}$
Two coefficients exceed the middle coefficient; their exact positions are not computed.
Bound on $\log_2 p_s$
$p_s = d_s/D(G,1)$
−80
−106
−150
−189
−200
some i in J_A
$p > 2^{-106}$
some k in J_C
$p > 2^{-106}$
Explicit middle index j
$p_j < 2^{-189}$
−1.09
−0.91
−0.86
−0.81
−0.70
Centered size $(s - K)/b$, where $K = 5b + L/2$
Inequality certificate, not an exact coefficient curve. Arrows indicate strict bounds.
Horizontal segments mark possible locations of the two existential coefficients.